

BARIS ASKIN

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EDUCATION

Carnegie Mellon University, Pittsburgh, USA

August 2022 - June 2027 (Anticipated)

· Ph.D. in Electrical and Computer Engineering

GPA: 4.0/4.0

Advisors: [Dr. Gauri Joshi](#) & [Dr. Carlee Joe-Wong](#)

· M.Sc. in Electrical and Computer Engineering (Aug. 2025, GPA: 4.0/4.0)

Bilkent University, Ankara, Turkey

September 2017 - June 2022

· B.Sc. in Electrical and Electronics Engineering

GPA: 3.99/4.0

Valedictorian – Highest Ranked Graduate of 2022 in the Faculty of Engineering

RESEARCH INTERESTS

Language Models, Federated Learning (FL), Distributed Optimization

WORK & RESEARCH EXPERIENCE

Graduate Researcher

August 2022 – Present

Carnegie Mellon University, Pittsburgh, PA

- Investigating how language models plan their outputs from an information-theoretic perspective.
- Working on generalization of misalignment in language models.
- Exploring language model routing systems in federated settings.
- Developing efficient fine-tuning methods for language models in distributed settings.
- Designing synchronous and asynchronous FL algorithms with communication- and time-efficiency and convergence guarantees.

Research Intern

May 2025 – August 2025

NVIDIA Corporation

- Worked on a scalable and asynchronous FL algorithm using data-free knowledge distillation [[Paper](#)]

Quant Researcher Intern

June 2026 – August 2026

IMC Trading

- Worked as a quant researcher intern at IMC Trading

Undergraduate Researcher

September 2020 – July 2022

[Imaging and Computational Neuroscience Lab](#), Bilkent University, Ankara, Turkey

- Worked on deep learning techniques for medical imaging under the supervision of Dr. Tolga Cukur.
- Worked on super-resolution of Magnetic Particle Imaging (MPI) System Matrices with deep learning methods to accelerate the calibration process.
- Worked on learning-based image reconstruction techniques for MPI. Proposed the first deep plug-and-play priors-based method for MPI reconstruction.

RELEVANT COURSES AND TOOLS

CMU: Advanced Introduction to Machine Learning, Convex Optimization, Intermediate Statistics, ABCDE of Statistical Methods in Machine Learning, Generative AI, Machine Learning with Large Datasets, Fundamentals of MDPs and RL, Deep RL

Bilkent University: Linear Algebra, Statistical Learning and Data Analytics, Stochastic Models, Medical Image Reconstruction and Processing, Probability and Statistics, Signals and Systems, Differential Equations, Digital Signal Processing, Telecommunications, Computer Networks

Tools: Python, MATLAB, Java, Assembly, VHDL, PyTorch, TensorFlow, OpenCV, Apache Spark

SELECTED PUBLICATIONS

1. **Internal Planning in Language Models: Characterizing Horizon and Branch Awareness**
M. Ustaomeroglu*, **B. Askin***, G. Joshi, C. Joe-Wong, G. Qu *equal contribution
International Conference on Learning Representations (ICLR), 2026 [\[Link\]](#)
2. **Emergent and Subliminal Misalignment Through the Lens of Data-Mediated Transfer**
B. Askin*, M. Ustaomeroglu*, A. Nayak*, G. Joshi, G. Qu, C. Joe-Wong
(Preprint, under review, 2026) [\[Link\]](#)
3. **Federate the Router: Learning LM Routers with Sparse and Decentralized Evaluations**
B. Askin*, S. Patel*, A. Nayak*, A. Vigano, J. Woo, G. Joshi, C. Joe-Wong
(Preprint, under review, 2026) [\[Link\]](#)
4. **Reviving Stale Updates: Data-Free Knowledge Distillation for Asynchronous FL**
B. Askin, H. R. Roth, Z. Sun, C. Joe-Wong, G. Joshi, Z. Xu
(Preprint, under review, 2026) [\[Link\]](#)
5. **PubSwap: Public-Data Off-Policy Coordination for Federated RLVR**
A. Nayak*, **B. Askin***, M. Ustaomeroglu, C. Joe-Wong, G. Joshi
(Preprint, under review, 2026) [\[Link\]](#)
6. **Federated Communication-Efficient Multi-Objective Optimization**
B. Askin, P. Sharma, G. Joshi, C. Joe-Wong
International Conference on Artificial Intelligence and Statistics (AISTATS), 2025 [\[Link\]](#) [\[Code\]](#) [\[Poster\]](#)
7. **Ravan: Multi-Head Low-Rank Adaptation for Federated Fine-Tuning**
A. Raje, **B. Askin**, D. Jhunjhunwala, G. Joshi
The Conference on Neural Information Processing Systems (NeurIPS), 2025 [\[Link\]](#)
8. **FedAST: Federated Asynchronous Simultaneous Training**
B. Askin, P. Sharma, C. Joe-Wong, G. Joshi
The Conference on Uncertainty in Artificial Intelligence (UAI), 2024 [\[Link\]](#) [\[Code\]](#) [\[Poster\]](#)
Selected papers on deep learning applications in medical imaging while I was at Bilkent:
9. **DEQ-MPI: A Deep Equilibrium Reconstruction with Learned Consistency for MPI**
A. Güngör, **B. Askin**, D. A. Soydan, C. B. Top, E. U. Saritas and T. Çukur
IEEE Transactions on Medical Imaging, Aug. 2023 [\[Link\]](#) [\[Code\]](#)
10. **TranSMS: Transformers for Super-Resolution Calibration in Magnetic Particle Imaging**
A. Güngör, **B. Askin**, D. A. Soydan, E. U. Saritas, C. B. Top and T. Çukur
IEEE Transactions on Medical Imaging, July 2022 [\[Link\]](#) [\[Code\]](#)

PROFESSIONAL SERVICE

- Reviewer for NeurIPS, ICLR, ICML, AISTATS, ISIT, AAAI, TMLR, IEEE/ACM ToN and MLSys
- Teaching Assistant for [Introduction to ML for Engineers \(18-661\)](#) in Spring 2024 and [Algorithms for Large-scale Distributed ML and Optimization \(18-667\)](#) in Fall 2024

SELECTED AWARDS AND ADDITIONAL INFORMATION

- 2023-2024 Ben Cook Presidential Graduate Fellowship in Electrical & Computer Engineering at CMU
- 2022-2023 Carnegie Institute of Technology Dean's Fellowship at CMU
- Bilkent University Comprehensive Scholarship: Full tuition waiver & stipend during the B.Sc. Program
- Scholarship of Turkish Ministry of Youth and Sports: Awarded stipend during the B.Sc. Program
- Ranked 3rd/104k in Academic Personnel and Postgraduate Education Entrance Exam in Turkey
- Ranked 252nd/2M in Nationwide University Entrance Exam in Turkey
- Volunteer at Young Guru Academy (an international NGO based in Istanbul) from 2018 to 2020